Most computer systems use the sRGB colour model for defining colours. This is a standardised coloration model jointly developed by Microsoft and Hewlett-Packard. The sRGB colour model guarantees that individual devices working on the RGB code (red, green, blue) reproduce colours logically and realistically, but the range of colours with this model is very restricted. When defining sRGB, the lowest common denominator is taken as the basis because the aim is to permit the rendition of colours on the medium that is most limited with respect to colour reproduction. So with conversion from YCbCr to sRGB, a multitude of colour shades are lost and this is why the printed image is not up to the mark. But with PIM technology, inkjet printers that are PIM enabled can now reproduce the colours in the YCbCr colour space.

Priming it with PIM

In PIM, the entire colour space that the camera is capable of writing is put in an EXIF header, which is a JPEG header that can contain camera-related information such as the definition of print gamma (gamma settings used by the printer) and the definition of the colour space actually captured by the camera. Further, camera information relevant to the printing process is also written into this header—the gamma value, the reference colour and the values for light and shade, colour saturation and colour balance, brightness, contrast and definition. As these values differ from camera to camera and from printer to printer, the original image (from the camera) often does not agree with the finished photo produced by the printer using conventional technology. This is where PIM technology comes handy—it recognises such discrepancies and the printer driver itself is able to interpret the supplementary image information contained in the EXIF header and it adjusts the print accordingly.

The print can also be adjusted for the subject. If a camera has the subject programming facility and is set for portrait photography, prints are produced with some amount of soft focus with optimised skin tones, while in the case of macro shots the priority is given to absolute definition and maximum contrast. Thus, the image capture process is optimised for the type of format selected.

With PIM technology, digital camera manufacturers can incorporate additional print-relevant data and commands in the EXIF header. Initially, only the Epson PhotoQuicker software will enable analysis of all the print-relevant data contained in the EXIF header and the conditioning of this data for printing. This software will be supplied with future PIM-compatible printers from Epson such as the Epson Stylus Photo 785EPX and allow camera images to be displayed, selected and printed.

Future developments in PIM technology will also support the DPOF (Digital print Order Form) specification—a kind of digital 'envelope'. This format of recording information allows images captured by digital cameras to be automatically printed through photo-finishing print services or home printers and allows pictures to be printed with pre-made frame patterns.

Epson has also included PIM into its digital-camera range such as in the Epson CP-920Z. It is presently working on a plan to license PIM technology to software manufacturers and printer manufacturers for a suitable fee so that image manipulation programs and printers can support this technology in the near future.

PIM technology brings a genuine quality advantage for users in practical use since it will allow home and amateur users to create prints that match those that were until now only seen in professional applications. d